

Mechanism Questions #3

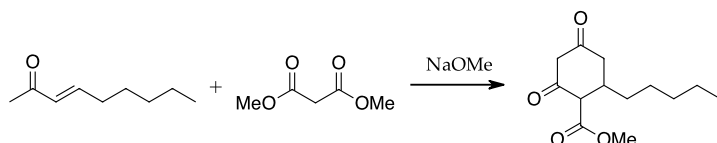
Barry Linkletter

This document was produced using the $\LaTeX$ typesetting language with the Tufte-handout document class. Chemical diagrams were prepared in *ChemDoodle*.

Explain the reaction mechanisms for the transformations below. Draw out each step and show electron movement using curved arrows. Identify the likely rate-determining step and identify the driving force for each reaction.

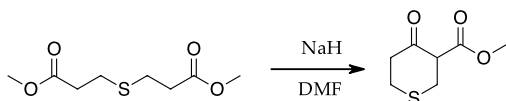
Each reaction is a single step in a multistep synthesis described in the referenced literature. For extra experience, access each paper and describe every step in the synthesis that is presented.

1.



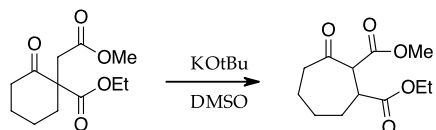
"A simple and efficient synthesis of 5'-($^2\text{H}_3$)olivetol." Michel Girard, David B. Moir, John W. ApSimon. *Can. J. Chem.*, 1987, 65, 189-190. <https://doi.org/10.1139/v87-029>

2.



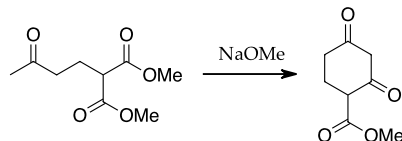
"Regioselective synthesis of cyclopentenones from 4-thianone." Haruo Matsuyama, Yasuyuki Miyazawa, Yuji Takei, and Michio Kobayashi, *J. Org. Chem.*, 1987, 52, 1703-1710. <https://doi.org/10.1021/jo00385a011>

3.



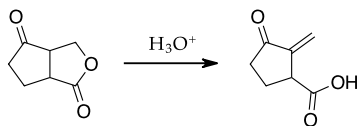
"A Facile Ring Enlargement." Zhuo-Feng Xie, Hiroshi Suemune, Kiyoshi Sakai, *Synth. Commun.*, 1989, 19, 987-992. <https://doi.org/10.1080/00397918908051019>. Available via ILL.

4.



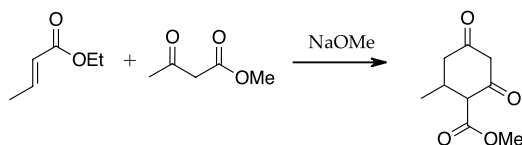
"Bridged Xanthenes. I. An Intermolecular Cycloaddition Route." Peter Yates and D. J. Bichan. *Can. J. Chem.*, 1975, 53, 2045-2053. <https://doi.org/10.1139/v75-285>

5.



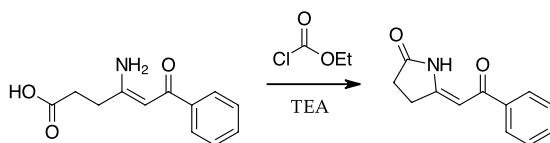
"Regiospecific synthesis of sarkomycin and some analogs." John N. Marx, Gevorg Minaskanian, *J. Org. Chem.*, 1982, 47, 3306-3310. <https://doi.org/10.1021/jo00138a020>

6.



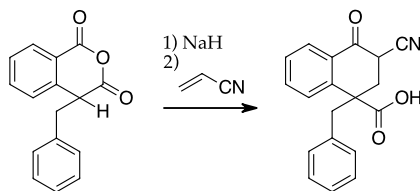
"Cyclization Reactions Leading to β -Hydroxyketo Esters." Jesse M. Nicholson, Ivan O. Edafigho, Jacqueline A. Moore, Vida A. Farrar, K.R. Scott, *J. Pharm. Sci.*, **1994**, *83*, 76-78. <https://doi.org/10.1002/jps.2600830118>

7.



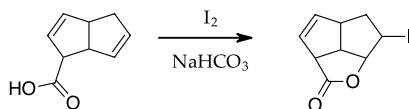
"Synthesis of 14-Hydroxy-8-Azaprostanoic Acids Methyl Ester." A. Barco, S. Benetti, G.P. Pollini, B. Veronesi, P.G. Baraldi, M. Guarneris, C.B. Vicentini, *Synth. Commun.*, **1978**, *8*, 219-226. <https://doi.org/10.1080/00397917808065612>. Available via ILL.

8.



"New bicyclic enamines and iminium salts. II. Synthesis of 1,4-dihydro-1,4-ethanoquinolinium salts and 4,5-dihydro-1H-1,4-methano-3-benzazepinium salts by reaction of bridged lactams with organometallic reagents." Gordon N. Walker, David Alkalay, *J. Org. Chem.*, **1971**, *36*, 491-500. <http://doi.org/10.1021/jo00803a001>

9.



"Bicyclo[3.2.1]octenones as Building Blocks in Natural Products Synthesis. 2. Formal Synthesis of ($\pm$)-Verbenalol." Chi-Peng Chang, Lung-Fu Hsu, Nein-Chen Chang, *J. Org. Chem.*, **1994**, *59*, 1898-1899. <https://doi.org/10.1021/jo0086a049>

10.



"Direct Synthesis Of γ -Butyrolactones Via γ -Phenyl Substituted Butyric Acids Mediated Benzyl Radical Cyclization." N.O. Mahmoodi, M. Jazayri, *Synth. Commun.*, **2001**, *31*, 1467-1475. <https://doi.org/10.1081/SCC-100104057>

Learning Goals

Through the suggested textbook reading, problems and participating in this class discussion, students will have...

- reviewed organic chemistry mechanisms featuring carbonyl groups and enols or enolates.